

## DynamoDB step-by-step guide

### Introduction

This document provides step-by-step instructions for creating and configuring the Amazon DynamoDB table required for the EcoStream telemetry data environment. The instruction demonstrates how to import the uploaded telemetry dataset in Amazon S3 into a DynamoDB table called “23122931-ecostream-telemetry-db”.

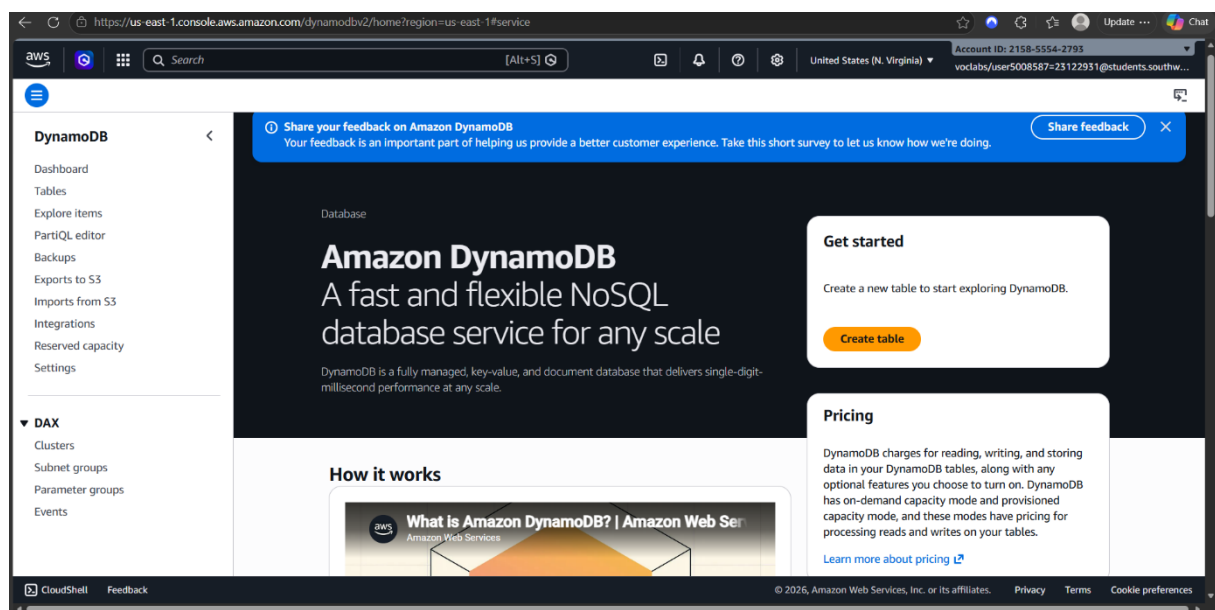
This document contains the parameters applied during the import process, such as choosing the partition key, table configuration settings, and import validation methods. Relevant screenshots and technical descriptions are provided throughout the guide to ensure that EcoStream workers can successfully reproduce the DynamoDB deployment and data import procedure in the AWS environment.

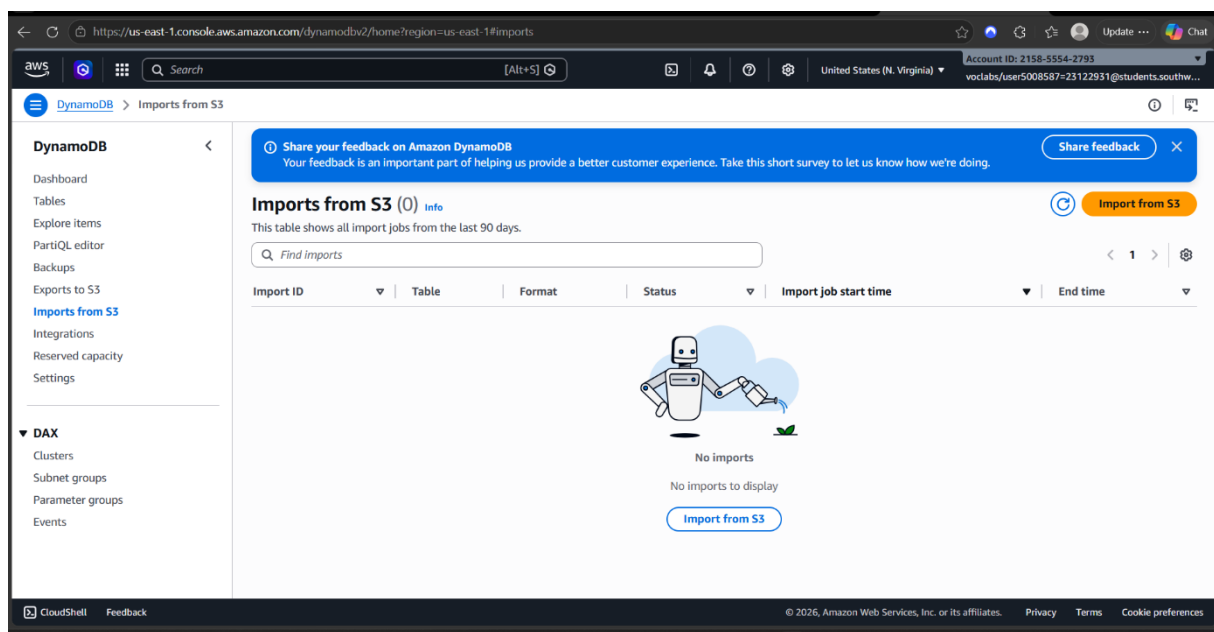
**Step 1:** Having completed the S3 bucket configuration and uploaded the dataset, the AWS search bar was used to locate for the Amazon DynamoDB service. DynamoDB was then picked to start the process of establishing the NoSQL database environment needed to import and query the EcoStream telemetry dataset stored in Amazon S3.

The DynamoDB dashboard was opened successfully, displaying the database management interface and available DynamoDB features. In the navigation panel, the “Imports from S3” option was selected to create a DynamoDB table directly from the dataset file stored in the previously created S3 bucket.

The DynamoDB dashboard was successfully opened, and the database administration interface and available functionalities of DynamoDB were displayed. In the navigation panel, the “Imports from S3” option was selected to create a DynamoDB table directly from the dataset file stored in the previously created S3 bucket.

This step was necessary because the assessment required importing the EcoStream telemetry dataset from Amazon S3 into a DynamoDB table for querying and analysis. The “Imports from S3” feature provided a direct integration between Amazon S3 and DynamoDB, simplifying dataset import.



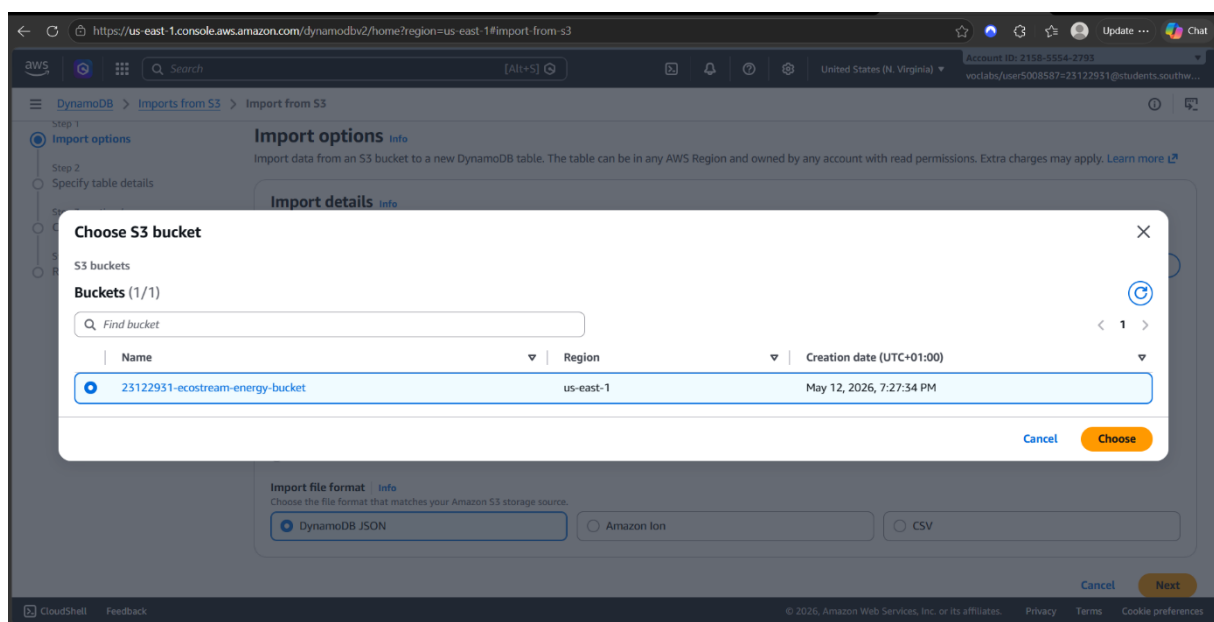


**Step 2:** The S3 bucket containing the EcoStream telemetry dataset was selected using the “Browse S3” option within the DynamoDB import configuration page. The bucket “23122931-ecostream-energy-bucket” was successfully identified and selected from the list of available S3 buckets in the AWS account.

The current AWS account was retained as the S3 bucket owner because both the S3 bucket and DynamoDB resources were created within the same AWS learner lab environment. The import file compression setting was left at “No compression” because the uploaded dataset file was stored in its original, uncompressed format. Under the import file format settings, the “DynamoDB JSON” option was selected because the uploaded telemetry dataset used the DynamoDB-compatible JSON structure required for direct table import into Amazon DynamoDB.

These settings ensured that DynamoDB could correctly locate, interpret, and import the EcoStream telemetry dataset from Amazon S3 into a new NoSQL database table for further querying and analysis tasks.

After configuring the “Import options”, the “Next” option was selected to continue to the “Specify table details” page.



https://us-east-1.console.aws.amazon.com/dynamodbv2/home?region=us-east-1#import-from-s3

Account ID: 2158-5554-2793  
voclabs/user5008587-23122931@students.southw...

**Import options** info

Import data from an S3 bucket to a new DynamoDB table. The table can be in any AWS Region and owned by any account with read permissions. Extra charges may apply. [Learn more](#)

**Import details** info

**Source S3 URL**  
Choose the Amazon S3 storage source to import from. You can specify a bucket or a path within the bucket.  
 [View](#) [Browse S3](#)  
 Example formats: s3://countries or s3://countries/territories/

**S3 bucket owner**  
☒ This AWS account (215855542793)  
☐ A different AWS account

**Import file compression**  
Choose the compression type that matches your source S3 data.  
☒ No compression  
☐ GZIP  
☐ ZSTD

**Import file format** info  
Choose the file format that matches your Amazon S3 storage source.  
☒ DynamoDB JSON ☐ Amazon Ion ☐ CSV

[Cancel](#) [Next](#)

**Step 3:** The DynamoDB table configuration page was completed by specifying the table details needed to import the EcoStream telemetry dataset from Amazon S3. The table was named “23122931-ecostream-telemetry-db” to clearly identify it as the main telemetry database table for the assessment task.

The partition key was configured as “Facility ID” with the data type set to “String”. This field was selected because each facility identifier in the dataset is unique to its registered region, making it suitable as the primary key for organizing and efficiently retrieving records in DynamoDB.

No sort key was configured because the assessment requirements only required a primary partition key for uniquely identifying records. After confirming the table configuration settings, the “Next” option was selected to continue to the table settings configuration stage.

This step was necessary because DynamoDB requires a primary key to organize and distribute data efficiently across the database infrastructure, enabling scalable querying and storage operations.

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voclabs/user5008587-23122931@students.southw...

**Specify table details**

**Table details** info  
DynamoDB is a schemaless database that requires only a table name and a primary key when you create the table.

**Table name**  
This will be used to identify your table.  
  
 Between 3 and 255 characters, containing only letters, numbers, underscores (\_), hyphens (-), and periods (.).

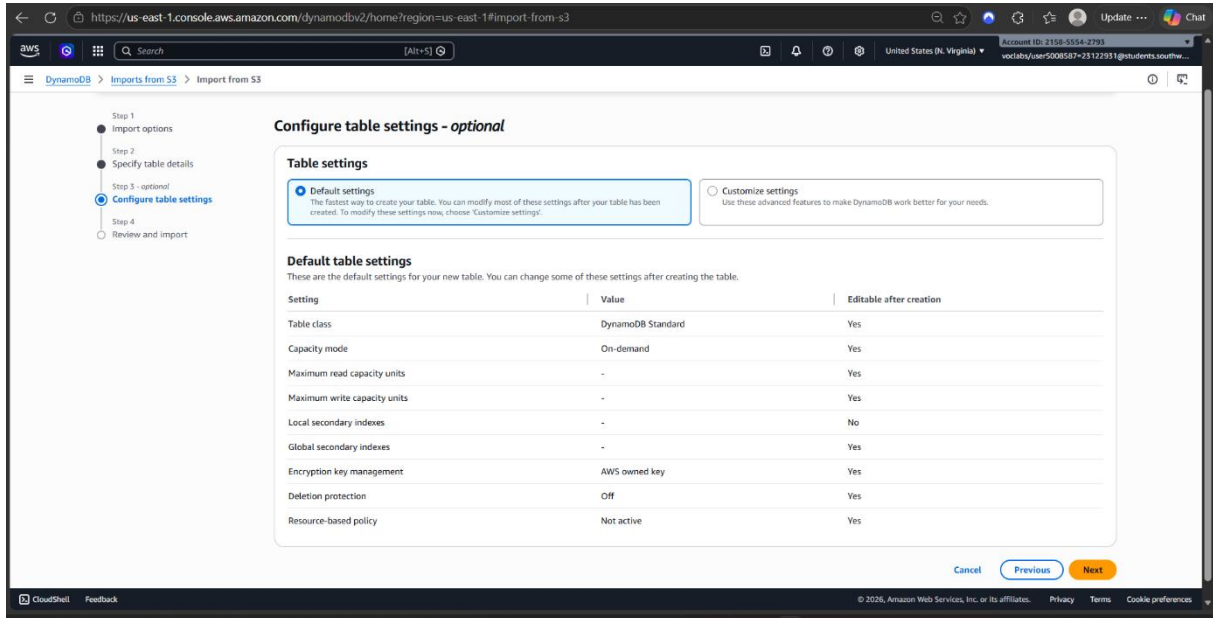
**Partition key**  
The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.  
   
 1 to 255 characters and case sensitive.

**Sort key - optional**  
You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.  
   
 1 to 255 characters and case sensitive.

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**Step 4:** The DynamoDB table settings configuration page was opened to review the default database configuration options before importing the telemetry dataset. The “Default settings” option was retained to simplify the deployment process and allow AWS to automatically apply the recommended DynamoDB configuration values.

This step was necessary because the default DynamoDB settings provided a fully managed, scalable NoSQL database configuration that efficiently stored and queried the EcoStream telemetry dataset without additional administrative overhead.



**Step 5:** The final review page was opened to verify all DynamoDB import configurations before starting the database import process. The review confirmed that the S3 source bucket “s3://23122931-ecostream-energy-bucket” was correctly selected and that the dataset format was configured to “DynamoDB JSON”.

The destination table details were also reviewed, confirming that the table name “23122931-ecostream-telemetry-db” and partition key “Facility ID” were correctly configured. The table class remained “Standard” with “On-demand” capacity mode enabled to allow automatic scaling of database read and write operations.

Additional settings, such as AWS-owned encryption keys and disabled deletion protection, were also verified before proceeding. After confirming that all configurations matched the assessment requirements, the “Import” button was selected to begin importing the telemetry dataset from the S3 bucket into DynamoDB.

This step was necessary to validate that the S3 source location, DynamoDB table structure, and database settings were correctly configured before creating the telemetry database.

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Step 1

Import options

Step 2

Specify table details

Step 3 - optional

Configure table settings

Step 4

Review and import

Review and import

Step 1: Import options

Edit

Import options

Source S3 URL

s3://23122931-ecostream-energy-bucket

S3 bucket owner

This AWS account

Import file format

DynamoDB JSON

Import compression type

No compression

Step 2: Destination table

Edit

Table details

Table name

23122931-ecostream-telemetry-db

Partition key

Facility ID

Table class

Standard

Sort key

-

Read/write capacity settings

Capacity mode

On-demand

Maximum read request units

-

Maximum write request units

-

Secondary indexes

Secondary indexes

No secondary indexes have been created.

Encryption at rest

Encryption key

AWS owned key

Deletion protection

Deletion protection

☐ Off

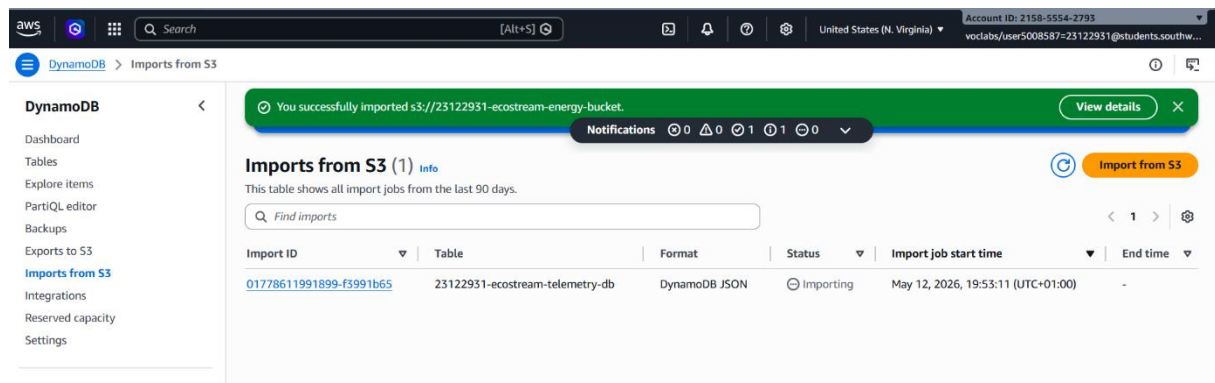
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Import

**Step 6:** The DynamoDB Imports from S3 dashboard confirmed that the import process for the EcoStream telemetry dataset had started successfully. The import job displayed the destination table “23122931-ecostream-telemetry-db,” the selected “DynamoDB JSON” file format, and the import status.

The import monitoring page confirmed that AWS was processing the telemetry dataset from the S3 bucket and automatically creating DynamoDB table records. This step verified that the integration between Amazon S3 and DynamoDB was functioning correctly and that the telemetry data import operation had been initiated successfully.



**Step 7:** The Amazon S3 bucket page was revisited to verify that the uploaded telemetry dataset “energy-telemetry-2025-2026-dbjson.json” remained successfully stored inside the “23122931-ecostream-energy-bucket”. The Objects section displayed the uploaded JSON dataset together with its file size, storage class, and last modified date.

This verification step confirmed that the source dataset remained available and accessible for DynamoDB import operations. It also confirmed that the telemetry file was correctly stored within the S3 bucket and ready for future querying, backup, or database integration tasks required by the assessment.

